



Energy-neutral networked wireless sensors



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ABSTRACT

Solar cells combined with power management algorithms enable the dynamic scheduling of Wireless Sensor Networks applications in a reference period, where the objective of the scheduling is to maximize the application quality level while conserving an energy level sufficient to constantly maintain the sensor operation. With the purpose of fulfillment of its activities the sensor may have alternative different applications with different costs and quality levels, which are intended to express how much the application meets the user requirements. In this paper we propose an algorithm that aims to find a (sub)optimal scheduling that maximizes the overall quality of service for two networked, solar cells powered sensors, and keeps the system energy neutral, thus ensuring that the system works uninterruptedly.

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1. Introduction

A Wireless Sensor Network (WSN) comprises a large number of low-power wireless embedded devices, called sensors, which can be easily deployed in a given area, in order to collect environmental data and detect the occurrence of events of interest. Once an event is detected, the sensors send any related sensed data to a special node, called sink, that acts as a gateway between the WSN and external networks and that forwards the sensed data to the user.

Most of the commercial sensors available nowadays are powered by limited-capacity batteries that provide the energy required to sustain the operation of the sensor by feeding its circuitry. Consequently, the sensor lifetime is also limited: when the battery drops below a minimum charge level, the sensor stops working. There is great emphasis on the efficient use of energy to prolong battery life [3,17,9]. Such a limitation may be addressed by means of energy harvesting systems that use the energy obtained naturally from the environment to power on the sensors. However, since the energy production of the energy harvesting systems is generally not continuous, as it depends on the conditions of the environment, the sensors generally use a battery that is recharged when there is an excess of energy production (i.e. it is produced more energy than required) by the energy harvesting system, and that guarantees the operation of the sensors when there is insufficient production of energy. In other words, the battery acts as a limited-capacity buffer of the energy. In such a system, an efficient energy management is necessary in order to guarantee an infinite sensor's lifetime while maintaining the activities of the sensors.

In this work we model the sensors activities as tasks; for example, a task may define sensing, processing, and communication activities with a given rate. Each task is associated to a cost (defined in terms of energy required by the task per unit of time) and to a Quality of Service (QoS) level, which expresses how much the task meets the user needs/requirements. We also model the energy production from a solar cell, which can be used to estimate the maximum amount of energy that the sensor could waste. Based on these models, we devise a scheduling strategy that consists in finding an assignment of execution tasks

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along a time frame, for instance one day, such that maximizes the QoS while keeping the sensor *energy neutral*, which means that within the time frame the energy production equals the energy consumption. Specifically, we address the problem considering a scenario that comprises two networked devices, one sensor and one sink, where both employ an energy harvesting system based on solar cells. Both the sensor and the sink perform a set of tasks that also include the communications among themselves. For this scenario we propose a low-complexity algorithm aimed at finding a (sub)optimal application scheduling that concerns both the sensor and the sink, such that the overall quality of the scheduling is maximized, the energy is preserved to guarantee the execution of applications even in conditions of darkness or low production, and the tasks selected at each time for the sensor and for the sink are compatible (i.e. they can work together). The optimization process is automatic and does not require human intervention.

In this paper we formulate the optimization problem for two networked sensors and present its evaluation. We have evaluated our approach by simulation and not by experimentation; note, however, that in order to construct more realistic models, both the energy consumption model from a sensor and the energy production model from a solar cell, are based on the parameters of real-world platforms: MicaZ and MSX-005F, respectively. Additionally, we discuss a generalization of the problem, which takes any number of sensors and a sink. It is important to point out that, even the scenario that we are addressing in this paper is limited to two nodes, to the best of our knowledge this is the first work that considers the optimization of the overall QoS of two networked sensors under the constraints of energy-neutrality and compatibility.

The remainder of this paper is organized as follows. After reviewing the related work in Section 2, we present in Section 3 our solar cell-based energy harvesting system, its energy production model and the energy consumption model of WSN applications. Section 4 describes our approach to schedule the applications to be executed on networked, solar cells powered wireless sensors, with the aim of optimizing their QoS. Section 5 provides the simulation results and Section 6 presents the generalization of the model for any number of sensors. Finally, Section 7 discusses the conclusions and further research.

2. Related work

Quality of Service (QoS) has been explicitly addressed under different perspectives in WSN. One of the most common approaches consists in looking for QoS at different layers of the network architecture, mainly MAC and routing layers. At MAC layer, several protocols [21] were proposed to improve the QoS based on service differentiation according to some static or dynamic criteria. At routing layer, QoS consists in finding the best path between a source and a destination node according to another metric, such as latency, bandwidth, packet loss, or fault tolerance. Some examples of these protocols are SAR [19], SPEED [11] and its evolution MMSPEED [10]. Ref. [20] discusses QoS at routing layer attending to the congestion control. QoS has been extensively studied at the scope of the application level, in such a way that the user is who may define application-specific performance metrics such as coverage, exposure, deployment, or reliability. In [13], for example, the optimum number of sensors that should send data is considered a QoS metric.

The sampling rate used by the application is another QoS metric with impact on the energy consumption of the sensor. Additionally, sampling could be provided simultaneously by different transducers able to monitor the same event but with different accuracy level and, in turn, different energy consumption. In this sense, several techniques focus on selecting the most appropriate transducer as well as its sampling rate according to the residual energy in order to balance sampling and energy [13,1]. Adaptive sampling techniques adjust the sampling rate of a sensor based on the spatial/temporal correlation of the data [1] to save energy. These techniques are often used in the sensors equipped with some energy harvesting system and able to estimate the energy that will be produced in a future time frame. For example, among the forms of energy harvested from the environment, it is known that the energy produced from the sun may be predicted with reasonable accuracy [2]. In [14] the authors present a theoretical model for planning and modeling micro-solar power systems. The model is validated by simulation and real experimentation, predicting with a minimal error the solar profile in different environments.

Several works combine adaptive sampling techniques and solar cells [18]. An example is [15], where the authors define the concept of *energy neutrality*, which states that the energy consumed by the sensor in a period of time is always less or equal to the energy harvested from the environment in the same period. Based on this, they propose an algorithm that divides the sensor lifetime in time slots, and that assigns to each slot the maximum sampling rate that keeps the system energy neutral. Although the authors do not mention explicitly QoS, we observe that increasing the sampling rate corresponds to rising the QoS of the application running on the sensor.

We have addressed the problem of QoS optimization while keeping the energy neutral condition in two previous works. Firstly, in [7], we propose an algorithm aimed at optimizing the QoS only attending to the duty cycle. We consider the re-optimization problem in [8], which considers an over/under production of energy with regard to the estimated production, and then find a (sub)optimal reassignment of applications to time slots that optimize the overall QoS while keeping the system energy neutral. Differently than the above works, in this paper we aim at the optimization of the QoS in a context of networked sensors, where the optimization should take into account also the compatibility of the scheduling plans of different devices and the fact that energy production and consumption is, in general, not the same on all devices. It is important to point out that, in this work, our view of QoS is rather different to the meaning used in the papers cited above. QoS enables to balance the application performance and its energy consumption, where performance is related to the use of different duty cycles, transducers and sampling rates, and/or communication patterns. To the best of our knowledge, this is the first work that consider the optimization of QoS in energy harvesting sensors under this perspective.

3. System model

This section describes the energy production model of a solar cell and the energy consumption model of a sensor. We recall that, as discussed in the Section 1, we consider an application scenario that comprises one sensor and one sink. Note, however, that this assumption does not prevent that the model can be applied to an arbitrary number of sensors. A sensor captures environmental data and transmits the results to the sink, which, in turn, stores and forwards the data to a remote user. We assume that both the sink and the sensors are powered by solar cells combined with embedded batteries. A device powered by means of a solar cell cannot count on a continuous power source (the energy production is dependent on the weather and time of the day), and uses the battery as an energy buffer to sustain its operations when there is no production of energy by the solar cells. If the battery level drops below a minimum threshold, the only alternative for the sensor is to switch to idle mode and wait until the solar cells start producing energy again.

In order to adapt to the user requirements, environmental situations, and available power, the application can adapt its duty cycles and monitoring tasks dynamically. Specifically, we assume that the application is structured in tasks corresponding to different monitoring activities and/or different duty cycles and different energy consumptions. Each task also corresponds to a different quality level of the monitoring tasks that should be maximized, while taking into account the drained energy, the residual power in the battery, and the expected energy production of the solar cells.

3.1. Solar cell energy production

We have designed an energy harvesting system that obtains energy from the sun light by means of solar cells, which transforms the solar energy into electricity. Thus, a device that is attached to a solar cell may use the electricity obtained from the solar cell to maintain their activities (under conditions of sufficient solar light) or obtain it from its batteries (under conditions of darkness or deficient sun light). In this subsection we present the requirements for the solar cell and for the device.

Solar cells are composed of wafers (in the order of micrometers thick) made of some semiconductor material at crystal-line state, typically silicon, with the property of converting the energy of the photons in electricity by the photovoltaic effect. The efficiency of a solar cell (η) is a measure of the energy conversion efficiency of the incident light absorbed by a solar cell into electric power. In turn, this depends on intrinsic factors of the region where it is located as, for instance, geographic coordinates, solar time, and inclination with respect to the sun rays. The efficiency is computed as $\eta = \frac{P_{out}}{D \times S}$, where P_{out} is the maximum output power provided by the cell (in W), D is the irradiance, defined as the density of incident power on a surface under standard assumptions (in W/m²) and S is the surface area of the solar cell (in m²). For this work we selected the MSX-005F solar cell [4] manufactured by BP-SOLAR, a polycrystalline cell with a Specified Load Voltage (V_{ld}) of 3.3 V, a P_{max} of 500 mW, a surface area (S) of 36 cm² and that provides a maximum current (I_{ld}) of 150 mA.

To compute the energy production of the solar cell we take into account the relevant parameters of the solar cell (η , S) and the average monthly solar irradiance (D) in some location. The output power is computed as $P_{out} = \eta \times D \times S$ and the average energy production E_p of the solar cell is P_{out} / V_{ld} . Thus, by taking the parameters D , η of a solar cell and the value of the irradiance D , we can obtain the energy production during a period of time. However, to obtain the instantaneous irradiance of the solar cell at time t , we should apply a correction factor depending on the current day and solar time. For example, in hours with no solar light the energy production is close to zero. Thus, at time t the instantaneous irradiance can be expressed as $D(t) = \frac{(t-h)^2}{4p} + D_{month}$, where t is any hour between 0:00 and 23:00, h is the hour of maximum solar light (12:00 am), D_{month} is equal to the average irradiance in a month and $p = 9/D_{month}$, where 9 corresponds to the number of standard hours of solar light, which depends on the month of the year. Fig. 1 shows the irradiance at time t represented as a quadratic function which draws a parabola with vertex at 12:00 in the noon. Then, the estimated energy production at time t is $E_p(t) = \frac{D(t) \times \eta \times S}{V_{ld}}$. In Fig. 1, the point (around 6:00) in which energy production begins corresponds to the sunrise and the point (around 18:00) in which energy production ends corresponds to the sunset.

We select MicaZ [5] as energy consumer device (both for the sensor and for the sink). MicaZ is a well-known platform for sensor networks applications, powered by 1.5 V 2-AA batteries with a capacity C (between 1500 and 3000 mA h), which should provide a voltage between 2.7 and 3.3 V for the correct working of the device. The battery discharge model depends mainly on the power requirements of the device. The battery charge model depends mainly on the solar cell energy production, which may lead to an increase of the stored energy. Thus, the time of charge (in h) that is required to completely charge a battery of capacity C , assuming no consumption and a constant energy production E_p is $\frac{C}{E_p}$ hours. Note that we used for this work the MSX-005F solar cell and the MicaZ platform, which provide the parameters for the production and consumption of energy, respectively; however, any other solar cell and sensor platform can be used instead.

3.2. Task model

We represent a sensor as an automaton that executes predefined tasks. In a given period of time, the sensor may be assigned to execute a subset of the available tasks $\Theta = \{T_0, \dots, T_a\}$. A task T_i ($i > 0$) is represented by a triple $T_i = \langle t_i, p_i, w_i \rangle$, where t_i is the execution time of the task, p_i is the period of execution of the task (note that must be $p_i > t_i$, otherwise the period of execution of the task cannot be met), and w_i is the energy consumption of the sensor due to the execution

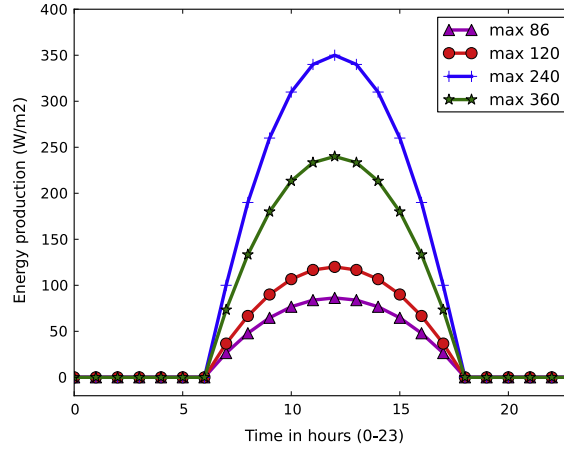


Fig. 1. Irradiance per hour for different D_{month} .

of task T_i . A special task T_0 (also called Idle task) is executed when there is no other task to execute, which means that T_0 does not have a period and a predefined execution time, and its energy consumption w_0 is defined as the energy consumed per unit of time. Hereafter we denote the execution of T_0 for a duration t as T_0^t .

A sensor executes a scheduling plan P that is defined as a finite sequence of tasks in Θ . Note that since the tasks are periodic, the scheduling plan P is cyclic and the total execution time of the cycle of P equals to the Least Common Multiple (LCM) of the period of the tasks, i.e. $t_p = LCM(T_0, \dots, T_x)$. The cost of execution of the scheduling plan P per unit of time is then given by:

$$c = \sum_{\substack{T_i \in \Theta \\ T_i \neq T_0}} \frac{w_i}{p_i} + w_0 \left(1 - \sum_{\substack{T_i \in \Theta \\ T_i \neq T_0}} \frac{t_i}{p_i} \right) \quad (1)$$

We consider the scenario described in Section 1, where a sensor interacts with a sink for communication purposes. Let us assume there are n possible scheduling plans of the sensor $P_1^N, \dots, P_n^N$ and m scheduling plans of the sink $P_1^S, \dots, P_m^S$. We use notation P_i^X , with $X \in \{S, N\}$, to denote the scheduling plan i that is either of the sensor (if $X = N$) or of the sink (if $X = S$). For a given scheduling plan P_i^X are defined its energy cost per unit of time c_i^X and a defined quality level q_i^X . Then, the usage of an energy harvesting system in the sensor and the sink restricts the set of scheduling plans that can be selected in both devices.

Let us assume two scheduling plans P_i^N and P_j^S with $i \leq n, j \leq m$. To enable the communication between the sensor and the sink, it exists a sending task $T_x \in P_i^N$ and a receiving task $T_y \in P_j^S$ ($T_x = \langle t_x, p_x, w_x \rangle, T_y = \langle t_y, p_y, w_y \rangle$). Then, P_i^N and P_j^S are compatible iff $(p_x) \geq (p_y)$ and $(p_x + t_x) \leq (p_y + t_y)$, which means that the time of transmission (in the sender) and the reception (in the receiver) must match. This is the purpose of many MAC protocols for WSN based on Low Power Listening [12,6,9]. In this paper we call to this condition compatibility among any pair of scheduling plans of the sensor and of the sink. We use a matrix C , composed by n rows and m columns, that represents the compatibilities between the scheduling plans of the sensor and of the sink by using simple binary notation: if $C(i, j) = 1$ then P_i^N and P_j^S are compatible scheduling plans; otherwise $C(i, j) = 0$, which means that the plans are incompatible and they cannot be run together.

A scheduling plan has associated a quality level q , which represents the degree of fulfillment of the user requirements. q is therefore a metric for QoS. Different scheduling plans of the sensor (or, equivalently, of the sink) may implement completely different functionalities or they may have the same functionalities although with different performance. For example, two different plans in the sensor P_i^N and P_j^N could perform the same set of tasks at different rates, e.g. sensing and forwarding at 4 and 60 s, respectively. Generally speaking, higher sampling rate implies higher quality level. Therefore, the quality level of P_i^N is larger than P_j^N . We observe that, without loss of generality, higher quality level implies larger cost. We denote as efficiency the ratio between the quality level and the cost ($\frac{q}{c}$). Note that quality level not only refers to the sampling rate or to the duty cycles but also to any feature that provides a better fulfillment degree of the activities in the device. For example, to perform the monitoring activity a sensor may use a different set of transducers, with different energy costs and different resolutions.

3.3. Energy model

Let B_{max}^X be the maximum battery capacity and B_{min}^X be the minimum battery capacity necessary to maintain the working of the device. A device connected to the solar energy harvesting system, as explained in Section 3.1, obtains the energy from

the sun to charge its batteries accordingly to the light conditions at each moment. Thus, the energy available in the batteries may increase (or decrease) depending on the solar irradiance in a place and instant of time.

As observed in [15], an energy harvesting sensor can operate uninterruptedly if its energy management is *energy neutral*. Considering a reference period of time $[t', t'']$ the energy management of a sensor is energy neutral if $B_{t'} = B_{t''}$. In our case, considering that the energy harvesting system is based on solar cells, the energy production has a cycle of 24 h, for this reason we aimed at devising an energy management strategy that is energy neutral in the 24 h. To this purpose we consider a time frame of 24 h divided into σ slots of duration $\delta = \frac{3600 \times 24}{\sigma}$ seconds.

Let $e_p^N(i)$ and $e_p^S(i)$ be the energy production at the end of slot i in the sensor and the sink, and $e_e^N(i)$ and $e_e^S(i)$ be the expected energy production (estimated by using the model in Section 3.1) at the end of slot i , of the sensor and the sink, respectively. Similarly, let $c^N(i)$ and $c^S(i)$ be the energy cost due to the execution of a scheduling plan during the slot i , in the sensor and the sink, respectively.

We denote with $B^X(i)$ the battery level at the end of slot i of the device X . Let also $\phi \in [0, 1)$ be the charging efficiency of the battery and the rectifier function $[x^X]^+$ be defined as:

$$[x^X]^+ = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (2)$$

If the duration of the slots is small enough we can assume that if the charge tends to overflow the battery above its maximum B_{max}^X , then the battery charge will be B_{max}^X also at the end of the slot. Similarly, if the charge tends to underflow the battery below 0, then the battery charge will be 0 also at the end of the slot. This assumption is motivated by the fact that the slots are short, the energy production will keep the same trend while the energy consumption will remain almost the same. Under these assumptions, and disregarding the power leakage of the battery, we can express the estimated battery level of the sensor and of the sink for each slot $i \in [1, \sigma]$ as follows:

$$B^X(i) = \max\{B_{max}^X, B^X(i-1) + \phi[e_e^X(i) - c^X(i)]^+ - [c^X(i) - e_e^X(i)]^+\} \quad (3)$$

where $B^X(0)$ is the initial battery level at the beginning of slot 1, the second term accounts for the energy which is produced and not consumed by X (which recharges the battery), and the last term accounts for the energy that is drained from the battery in X . This equation considers the fact that the battery is not an ideal energy buffer and it assumes that the excess energy that is neither used nor accumulated in the battery is dissipated by the system.

4. Energy neutral adaptivity

Although uncontrollable, the energy production given by a solar cell is predictable [15]. Based on this fact we devise a task scheduling strategy for the sensor and the sink aimed at achieving energy neutrality strategy of the overall systems.

We assume that the scheduling plans (and their respective costs and quality levels) for both the sensor and the sink are known, and that they are energy neutral at least in some condition. For example, a scheduling plan may not be energy neutral if executed in winter, but it may be energy neutral in summer (where the solar cells guarantee a larger energy production). Furthermore, as observed in Section 3.2, a scheduling plan of the sensor may be compatible only with some scheduling plans of the sink and vice versa. On the other hand, we assume that there always exists compatible scheduling plans to be executed by the sensor and the sink, and that the scheduling plans that keep in idle the sensor or the sink are always compatible.

The objective of the power management algorithm that we are proposing is to find an assignment of scheduling plans within a time frame such that maximizes the overall quality level, while maintaining the overall system energy neutral, and while guaranteeing the compatibility of the scheduling plans executed by the sensor and the sink. Let us remind that we consider a time frame of 24 h divided into σ slots. The power management algorithm returns two vectors S^N and S^S , that contain the allocation of the scheduling plans to each time slot for the sensor and for the sink, respectively. Specifically, $S^N[i] = j$ iff the algorithm assigns at the slot $i \in [1, \sigma]$ the scheduling plan P_j^N to the sensor, and $S^S[i] = k$ iff the algorithm assigns at the slot i the scheduling plan P_k^S to the sink.

For a given assignment $S^N[]$ and $S^S[]$, the energy budget of the sensor is $B^N(\sigma) - B^N(0)$ and of the sink is $B^S(\sigma) - B^S(0)$. We also define the vector q of the quality level, the vector g of the compatibility of the scheduling plans, and the two vectors c^N and c^S of the energy costs for the sensor and for the sink, respectively. Specifically, given the assignment S^N and S^S , for a given time slot $i \in [1, \sigma]$ let $S^N[i] = j$ and $S^S[i] = k$, we have that $q[i] = \min\{q_j^N, q_k^S\}$, where q_j^N and q_k^S are the quality of the plan j and k assigned for the sensor and for the sink, respectively at slot i ; $g[i] = C(j, k)$; $c^N[i] = c_j^N$ and $c^S[i] = c_k^S$.

Our algorithm computes the Quality of Service (QoS) for the sensor and for the sink as the sum of the quality levels of the scheduling plans assigned to all the slots. By maximizing the quality levels of the plans assigned we maximize, in turn,

the overall quality of service in the period of time considered. Thus, the problem of assignment can be formalized as an optimization problem as follows:

$$\begin{aligned}
 & \max \sum_{i=1}^{\sigma} q[i] \\
 & B^N(0) \leq B^N(\sigma) \wedge B^S(0) \leq B^S(\sigma) \\
 & B_{min}^N \leq B^N(i) \wedge B_{min}^S \leq B^S(i) \quad \forall i \in [1, \sigma] \\
 & \prod_{i=1}^{\sigma} g[i] = 1
 \end{aligned} \tag{4}$$

The first constraint of the above optimization problem state that the assignment of the scheduling plans must be energy neutral. The second constraint state that the battery level of the devices can never be below B_{min}^X because otherwise the sensor or the sink would have to stop their operations. Note also that the expression of $B^X(i)$ is not linear, thus leading to a non-linear optimization problem. Finally, the last constraint means that at each slot, the scheduling plans assigned to the sensor and to the sink must be compatible.

4.1. Power management algorithm

Let Φ^X be a set of scheduling plans (either of the sensor or of the sink). We define $opt(\Phi^X)$ as the scheduling plan $P_i^X \in \Phi^X$ such that $\forall P_j^X \in \Phi^X$ holds $\frac{q_i^X}{c_i^X} \geq \frac{q_j^X}{c_j^X}$ and $q_i^X > q_j^X$ if $\frac{q_i^X}{c_i^X} = \frac{q_j^X}{c_j^X}$ (i.e. P_i^X is the most efficient scheduling plan in Φ^X). This means that the function $opt(\Phi^X)$ returns the most efficient plan among a set of plans available for X . It is important to point out that we are sorting the plans by their efficiency in descendant order and not by their quality levels; efficiency allows us to balance quality and cost and to find progressively the next less efficient plan (but with higher quality level) or, alternatively, the next more efficient plan (but with lower quality level) that keep the system energy neutral. However, find the next plan with higher (or lower) quality level than the quality level of the current plan could converge into a non-energy-neutral system immediately in the next step. It should be also noted that a plan P_i^X for which $q_j^X < q_i^X$ and $c_j^X \geq c_i^X$ for some i is inefficient, and it can be excluded a priori. Also, we define the set $\Phi_q^X = \{P_i^X \in \Phi^X : q_i^X > q\}$ as the set of scheduling plans in Φ^X that have a quality level above q , and $\Phi_c^X = \{P_i^X \in \Phi^X : c_i^X < c\}$ as the set of scheduling plans in Φ^X that have a cost below c . Given a scheduling plan P_i^N for the sensor and a set of scheduling plans for the sink Φ , we define the subset of scheduling plans in Φ compatible with P_i^N as $comp(\Phi, P_i^N) = \{P_j^S \in \Phi : C(i, j) = 1\}$. Finally, we denote with r_1 and r_2 the slots immediately after the sunrise and the sunset, respectively.

We consider a setting in which the scheduling of the plans is executed periodically by the sink (or, equivalently, by the sensor) at the beginning of a 24 h period, in order to assign to both devices the scheduling plan for each time slot. In order to make this assignment, the sink should solve the optimization problem of Eq. (4). Since an exact solution of the problem requires computational resources generally not available in the considered class of devices, we propose a simple heuristic aimed at finding an admissible solution that approximates the optimal solution. In order to simplify the problem, we make the following assumptions:

Assumption 1. Given the scheduling plans P_i^N and P_j^N , let us consider two scheduling plans for the sink P_h^S, P_k^S such that P_h^S is the minimum cost scheduling plan for the sink in the set $comp(\Phi^S, P_i^N)$, and P_k^S is the minimum cost scheduling plan for the sink in the set $comp(\Phi^S, P_j^N)$. Then, $c_i^N \leq c_j^N$ iff $c_h^S \leq c_k^S$.

The previous assumption means that, if we execute a “low cost” scheduling plan for the sensor, it is possible to find a compatible, low cost scheduling plan also for the sink. Although this assumption limits the applicability of our heuristic, we note that this behavior is rather typical and captures a meaningful class of applications. For instance, consider the case in which the sensor has the option to sample the environment with two different rates, say 1 and 10 s, corresponding to two different scheduling plans (where the first scheduling plan costs more than the second). In the first case the sensor will send data to the sink every 1 s and then the sink should be ready to receive every 1 s, while in the second case the sink may reduce its duty cycle to receive data every 10 s. The result is that the sink should use a more costly scheduling plan in the first case and a less costly plan in the second case. This kind of applications fit the Assumption 1. Similarly, we also make the following assumption for the quality level of the scheduling plans:

Assumption 2. Given the scheduling plan P_i^N and P_j^N , let us consider two scheduling plans for the sink P_h^S, P_k^S such that P_h^S is the maximum quality level scheduling plan for the sink in the set $comp(\Phi^S, P_i^N)$, and P_k^S is the maximum quality level scheduling plan for the sink in the set $comp(\Phi^S, P_j^N)$. Then, $q_i^N \leq q_j^N$ iff $q_h^S \leq q_k^S$.

Algorithm 1. Assignment Algorithm

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1: Let  $P = \text{opt}(\Phi^N)$  and let  $q, c$  be the quality level and cost of  $P$ .
2: Let  $S^N[i] = P; S^S[i] = \text{opt}(\text{comp}(\Phi^S, P)) \forall i \in [1, \sigma]$ 
3: if  $(B^N(\sigma) \geq B^N(0) \wedge B^S(\sigma) \geq B^S(0)) \wedge (B_{\min}^N \leq B^N(i) \wedge B_{\min}^S \leq B^S(i)) \forall i \in [1, \sigma]$  then
4:   % improves the quality of the solution
5:   loop
6:      $\Phi^N = \Phi_q^N$ 
7:     Let  $P = \text{opt}(\Phi^N)$ , and let  $q, c$  be the quality level and cost of  $P$ .
8:      $\text{improved} = \text{Upgrade}(P)$ 
9:   while  $\Phi^N == \emptyset$  or  $\text{improved} == \text{FALSE}$ 
10: else
11:   % reduces the cost of the solution
12:   loop
13:      $\Phi^N = \Phi_c^N$ 
14:     Let  $P = \text{opt}(\Phi^N)$ , and let  $q, c$  be the quality level and cost of  $P$ 
15:      $\text{improved} = \text{Downgrade}(P)$ 
16:   while  $\Phi^N == \emptyset$  or  $\text{improved} == \text{FALSE}$ 

```

The pseudocode for the assignment of the scheduling plans to the time slots is presented in Algorithm 1, which progressively achieves a (sub)optimal solution in several iterations. Initially, it starts assigning to all the slots of the sensor the most efficient scheduling plan, and by finding, among the compatible schedules of the sink, the most efficient one (line 2). After this preliminary step, if the solution is admissible, then the algorithm enters a loop in which it tries to improve it (line 5). Otherwise, the solution is not admissible because it is not energy neutral or because the battery level of the sensor or of the sink drops below the minimum. In this second case the algorithm enters a loop (line 12) in which it looks for solutions that cost less (but that may have also a lower QoS).

In the first case (the initial solution is admissible), the algorithm enters a loop in which it considers only the scheduling plans of the sensor that have a better quality level than that of the scheduling plan used for the initial assignment. It selects among these the most efficient and tries to reassign to some slots this new scheduling plan. This reassignment is achieved by the Upgrade function (in line 8), shown in Algorithm 2. This loop ends only when there are no more better scheduling plans to try. Function Upgrade reassigns slot by slot (starting from the slot at sunrise, in which the solar cells start producing energy again) the scheduling plan for the sensor and, for each reassignment, it also finds a (new) compatible scheduling plan for the sink that, according to Assumption 2, must have a better quality level, thus this reassignment improves the overall quality level of the solution.

Algorithm 2. Upgrade Function

Require: P : a scheduling plan for the sensor.

Ensure: improved : a boolean value that states whether S^S and S^N had been improved

```

1: Let  $c_1$  be the cost of  $P$ 
2:  $\text{improved} = \text{FALSE}$ 
3: Let  $Q = \text{opt}(\text{comp}(\Phi^S, P))$ , and  $c_2$  be the cost of  $Q$ 
4:  $k = r_1; j = 1$ 
5: while  $(B^N(\sigma) - B^N(0) - c_1 + c^N(k) \geq 0) \wedge (B^S(\sigma) - B^S(0) - c_2 + c^S(k) \geq 0) \wedge (j \leq \sigma)$  do
6:    $H_1 = S^N[k]; S^N[k] = P$ 
7:    $H_2 = S^S[k]; S^S[k] = Q$ 
8:   if  $(B^N(i) < B_{\min}^N) \vee (B^S(i) < B_{\min}^S)$  for some  $i$  then
9:      $S^N[k] = H_1; S^S[k] = H_2$  % undo the replacement and exit
10:   return  $\text{improved}$ 
11:    $k = (k + 1) \bmod \sigma; j++$ 
12:    $\text{improved} = \text{TRUE}$ 
13: end while

```

In the second case (the initial solution is not admissible), the algorithm enters a loop in which it considers only the scheduling plans of the sensor that have a smaller cost than that of the scheduling plan used for the initial assignment. It selects among these the most efficient and tries to reassign to some slots this new scheduling plan. This reassignment is achieved by the Downgrade function (in line 15), shown in Algorithm 3. This loop ends only when there are no more scheduling plans with a lower cost to try. Similarly to Upgrade, Function *Downgrade* reassigns slot by slot (starting from the slot at sunset, in which the solar cells stop producing energy) the scheduling plan for the sensor and, for each reassignment, it also finds a (new) compatible, scheduling plan for the sink that, according to Assumption 1, must have a lower cost, reducing thus the overall cost of the solution.

Algorithm 3. Downgrade Function

Require: P : a scheduling plan.
Ensure: *improved*: a boolean value that states weather S^S and S^N had been improved

```

1: Let  $c_1$  be the cost of  $P$ 
2: improved = FALSE
3: Let  $Q = \text{opt}(\text{comp}(\Phi^S, P))$ , and  $c_2$  be the cost of  $Q$ 
4:  $k = r_2; j = 1$ 
5: while  $((B^N(\sigma) - B^N(0) < 0) \vee (B^N(i) < B_{\min}^N)) \wedge ((B^S(\sigma) - B^S(0) < 0) \vee (B^S(i) < B_{\min}^S))$  for some  $i \wedge (j \leq \sigma)$  do
6:    $S^N[k] = P; S^S[k] = Q$ 
7:   improved = TRUE
8:    $k = (k + 1) \bmod \sigma; j++$ 
9: end while
10: return improved

```

It can be observed that the complexity of Algorithm 1 is $O(n \cdot (\sigma + \max\{n, m\}))$. In fact, in each iteration of the main loop of Algorithm 1 (either the loop that increases the quality of the solution or the loop that decreases its cost) a scheduling plan P_i^N is selected and assigned to each one of the σ slots of the sensor by the Function Upgrade (or Downgrade). These functions take complexity $O(\sigma)$. When a plan P_i^N is selected, it is also removed from Φ^N (along with all the scheduling plans with a lower quality level or higher cost, depending on the case), thus reducing the number of scheduling plans in Φ^N of at least 1 element. As a consequence the main loop is repeated at most n times. Note that the removal of elements from Φ^N can be done in $O(n)$. Furthermore, each time a scheduling plan is reassigned to a slot in S^N , the loop also chooses a compatible scheduling plan to reassign to the same slot in S^S . The selection of a compatible scheduling plan can be done in $O(m)$. Consequently, the overall complexity is $O(n \cdot (\sigma + \max\{n, m\}))$. Considering that, typically $\sigma \gg n, m$ holds, the complexity reduces to $O(n \cdot \sigma)$.

4.2. Communication model

The communication between the sensor and the sink is based on a distributed model where the participants, a sensor and a sink, interchange data in order to implement the algorithms described in the previous subsection. **When the initial plan P is selected in the sensor, which occurs in the line 1 in Algorithm 1, the first interchange of messages takes place: the sensor sends P to the sink, which must select the most efficient plan that is compatible with P and assign it to all the slots.** The second interchange of messages occurs after the assignment and before evaluating the condition in Algorithm 1 (line 3). **This message is sent from the sink towards the sensor to communicate the result (true or false) of evaluating the condition of energy neutrality in the sink (i.e. $B^S(\sigma) \geq B^S(0) \wedge B_{\min}^S \leq B^S(i) \forall i \in [1, \sigma]$).** Based on the information received from the sink and its own information the sensor decides between improving the solution (line 5) or reducing the cost of the solution (line 12) and goes into the corresponding loop. The third interchange of messages occurs after line 7 or, alternatively, after line 14, to communicate to the sink the new plan selected in the sensor P as well as an additional field to indicate if doing Upgrade or Downgrade. Once received this message by the sink, both devices proceed individually with the execution of the corresponding function (Upgrade or Downgrade) until one of the next two things happen in the sink: **(1) σ slots were replaced with the new plan; or (2) a number i ($i < \sigma$) of slots were replaced (i.e. the assignment of the plan to the slot $i + 1$ becomes non-energy neutral the solution).** In any of the two cases the sink sends a message towards the sensor with the value of the variable *improved* and the value of the last slot assigned i exactly before line 10 or 12 (depending on the case) if it executes the function Upgrade, or alternatively, before line 7 or 11 (depending on the case) if it executes the function Downgrade. Fig. 2 represents the communication protocol to be implemented in the sensor and in the sink.

The information described above is the only data interchanged between the sensor and the sink to execute one iteration of the loop. If the solution can be still improved (or it is not energy-neutral) then the sensor starts a new iteration of the loop

msc Communication Protocol 1 Sensor-1 Sink

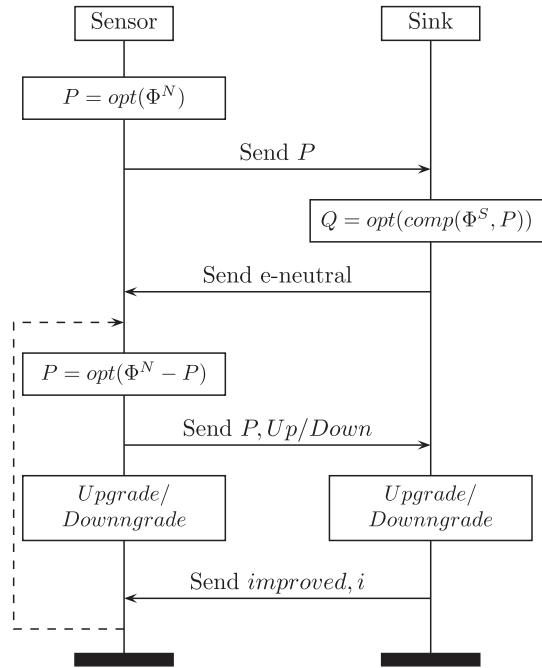


Fig. 2. Communication protocol between 1 sensor and 1 sink.

in line 5 (or line 12) of Algorithm 1 and proceeds as explained above. Thus, the maximum number of messages interchanged between the sensor and the sink for each execution of the optimization algorithm would be $2(n + 1)$, where n is the number of scheduling plans in the sensor.

5. Evaluation

We have evaluated our approach by using a simulator implemented in C. As described in Section 3.1, we simulate the energy production of the solar cell MSX-005F and the energy consumption of the sensor node platform MicaZ. The sensor and the sink can select different execution plans depending on the energy production and consumption in the sensor and the plans that are compatible in the sink. We consider a time frame of 24 h divided in 300 slots ($\sigma = 300$) with a duration of 288 s each one ($\delta = 288$).

To compute the energy production we take the data corresponding to the average monthly solar irradiance D_{month} in Madrid (data taken from RetScreen [16]): 84.6, 123.3, 178.8, 212.9, 247.9, 295.4, 300, 264.1, 202.9, 130.4, 88.8, 70.8 W/m², in the months of the year from January to December, respectively. For the sake of simplicity, in the evaluation we assume the same values of irradiance for the sensor and for the sink and, consequently, an energy production identical in both (note however that this is not a requirement of our algorithm). We compute the instantaneous energy production at any hour t ($E_p(t)$) by using the energy production model described in Section 3.1 and then we distribute $E_p(t)$ equally among the slots in an hour ($\frac{\sigma}{24}$). We have adapted the number of hours of solar light accordingly to each month varying the sunrise slot r_1 and the sunset slot r_2 in order to do more realistic simulations. We fix the maximum capacity of the battery B_{max}^x to be 2500 mA h and the

Table 1

Scheduling plans for the sensor employed in the simulation.

Plan	T_1	T_2	T_3	T_4	T_0	c	q (%)	Efficiency
N1	(2, 5, 10)	(1, 5, 15)	(1, 5, 8)	–	T_0^1	7.0	100	14.28
N2	(1, 10, 3)	(1, 10, 5)	(1, 10, 5)	(1, 10, 7)	T_0^6	3.2	90	28.12
N3	(5, 15, 10)	–	–	–	T_0^{10}	2.0	85	42.50
N4	(2, 20, 3)	(4, 20, 4)	–	–	T_0^{14}	1.75	80	45.71
N5	(20, 30, 10)	–	–	–	T_0^{10}	1.0	75	75
N6	(26, 30, 1)	–	–	–	T_0^1	0.3	70	233.3

Table 2

Scheduling plans for the sink employed in the simulation.

Plan	T_1	T_2	T_3	T_4	T_0	c	q (%)	Efficiency
S1	(1, 3, 2)	(1, 3, 10)	(1, 3, 12)	–	T_0^0	8.0	100.0	12.5
S2	(1, 5, 10)	–	–	–	T_0^0	3.6	98.0	27.2
S3	(1, 6, 7)	(1, 6, 3)	(1, 6, 5)	(2, 6, 2)	T_0^1	3.16	95.0	30.0
S4	(2, 10, 3)	(6, 10, 10)	–	–	T_0^2	1.7	90.0	52.94
S5	(7, 10, 10)	–	–	–	T_0^3	1.6	85.0	53.13
S6	(1, 20, 2)	(1, 20, 1)	(1, 20, 3)	(10, 20, 1)	T_0^7	1.05	80.0	76.19
S7	(27, 30, 7)	(2, 30, 10)	–	–	T_0^1	0.63	60.0	94.73
S8	(32, 35, 1)	–	–	–	T_0^3	0.2	50.0	250.0

Table 3

Compatibility matrix.

%	S1		S2		S3		S4		S5		S6		S7		S8	
	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
N1	0	1	0	1	1	1	0	1	0	0	1	1	0	1	1	0
N2	1	1	0	1	1	1	0	1	1	1	0	1	0	0	0	0
N3	0	1	1	1	1	0	0	1	1	1	0	0	0	1	0	1
N4	1	1	0	1	1	1	1	0	0	0	0	1	0	1	0	1
N5	0	0	0	1	1	1	1	1	0	1	1	1	0	1	0	0
N6	0	1	0	1	0	0	1	1	0	0	1	1	0	1	1	1

Legend: L: Low compatibility-40%. H: High compatibility-80%.

minimum B_{min}^x to be 1950 mA h; the battery level at slot 0 is 2000 mA h. To validate that the overall system is energy neutral we need to check the condition $B^N(300) \geq B^N(0)$ and $B^S(300) \geq B^S(0)$, and then proceed accordingly to Algorithm 1.

We define six scheduling plans for the sensor N1, ..., N6 and eight scheduling plans for the sink S1, ..., S8, whose description is detailed in Table 1 and 2 respectively. For each plan, the columns from 2 to 5 present the definition of the tasks (the notation “–” means that the plan does not contain that task); the column 6 indicates the execution time for the Idle task¹; and in the third last columns we specify the cost of the plan per unit of time c , its quality level q (in percentage), and its efficiency computed as $\frac{q}{c}$, respectively. Thus, the order of the plans by efficiency is {N6, N5, N4, N3, N2, N1} in the sensor and {S8, S7, S6, S5, S4, S3, S2, S1} in the sink. These plans were selected because they are representative of WSN applications which contain a different number of tasks (or activities) with different execution times, delivering different quality levels with a cost, where higher quality level involves larger costs.

Not all the plans in the sensor can run together with any plan in the sink. We refer this concept as compatibility, which we described in Section 3.2. To express the compatibility degree between a set of plans of the sensor and of the sink, we use a percentage, which is used to randomly generate the compatibility matrix. Specifically, we use two compatibility percentages, 40% and 80%, to represent that at most the 40% (low compatibility) and 80% (high compatibility) of the plans in the sensor are compatible with the plans in the sink, respectively. Table 3 shows the matrix representing a compatibility degree of at most 40% and 80% of the plans in the sensor and in the sink. We execute the initial assignment algorithm for the first day of each month of the year, starting at 0:00 am that corresponds to the slot 0.

We obtain an overall energy neutral system for the two percentages of compatibility considered for all months of the year, which means that the algorithm finds an assignment of scheduling plans to slots in the sensor and in the sink that verifies $B^N(300) \geq B^N(0)$ and $B^S(300) \geq B^S(0)$. We illustrate this in Fig. 3 for the sensor and in Fig. 4 for the sink, where the battery level for each slot is shown for each month of the year, for a compatibility of 40%. As observed, there is a clear relationship between the energy production and the battery level: the bigger is the production, the larger is the battery level achieved at the last light hour (close slot r_2) and the lower is the battery level before r_1 , due to the fact that the most consuming plans can be assigned when the production is larger. Both in the sensor and in the sink, the larger battery level is achieved in July (achieving 2015 mA h and 2030 mA h at slot r_2 , respectively). Correspondingly, the lowest battery level is achieved (close slot r_1) in the same month in the sensor (the most consuming plans are assigned also in that month) while in the sink is achieved in February due to Algorithm 1 assigns the optimal plan that is compatible with the plan in the sensor, which not necessarily is the plan with higher consumption.

We observe the impact of the compatibility on the sink in Fig. 4. As shown, the battery level at slot 300 is rather larger than that at the same slot of the sensor, which means that a part of the energy production is accumulated in the battery and it is not used to execute a most consuming plan with a highest quality of service. This effect occurs during the months of bigger production (from May to September). Remind that the sink executes the first most efficient plan while compatible

¹ Remind that T_0^t means that the duration of T_0 is t units of time.

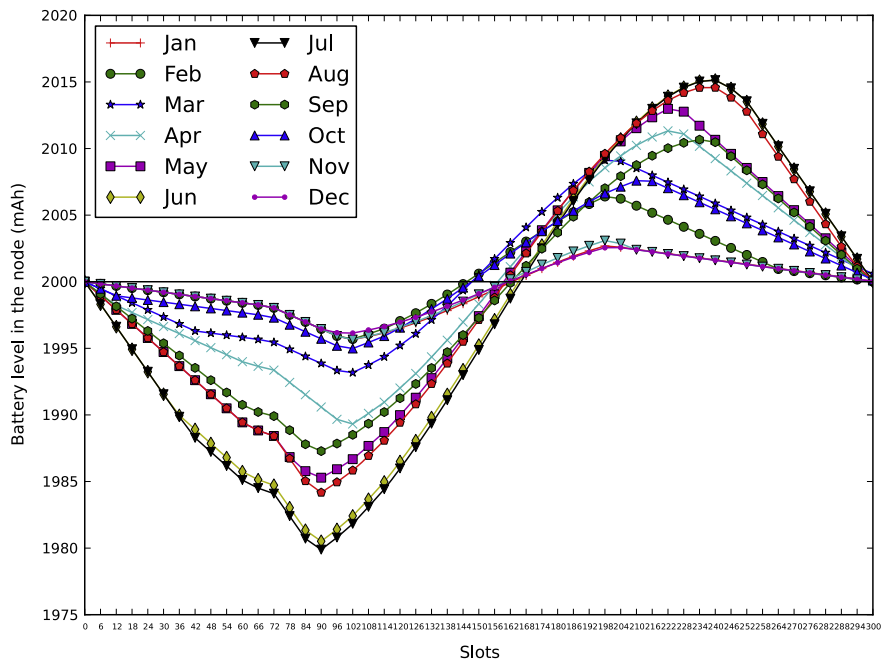


Fig. 3. Battery level in the sensor during the 1st day of each month of the year with a compatibility of 40%.

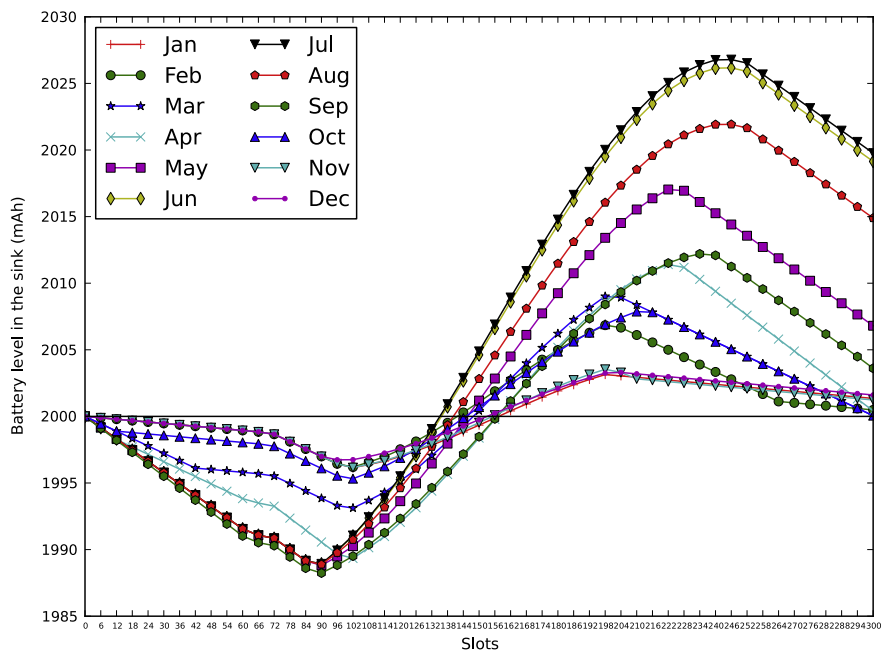


Fig. 4. Battery level in the sink during the 1st day of each month of the year with a compatibility of 40%.

with the plan in the sensor. Therefore, in spite of the sensor may change its scheduling plan, if the plan of the sink is still compatible with the new selected plan in the sensor, the sink continues executing the same plan, which leads to a biggest production while the consumption is kept. This is a direct consequence of the cooperation between nodes: while one of them could waste more energy if it executes alone, executing together involves that one of them must restrict to the other for the sake of achieving a maximum overall quality level. If we reduce the compatibility percentage we reduce, in turn, the probability of that a plan in the sensor be compatible with the most efficient plan in the sink, therefore, forcing to the sink to select another plan that is less efficient but has a higher consumption and a higher quality. This effect was checked by

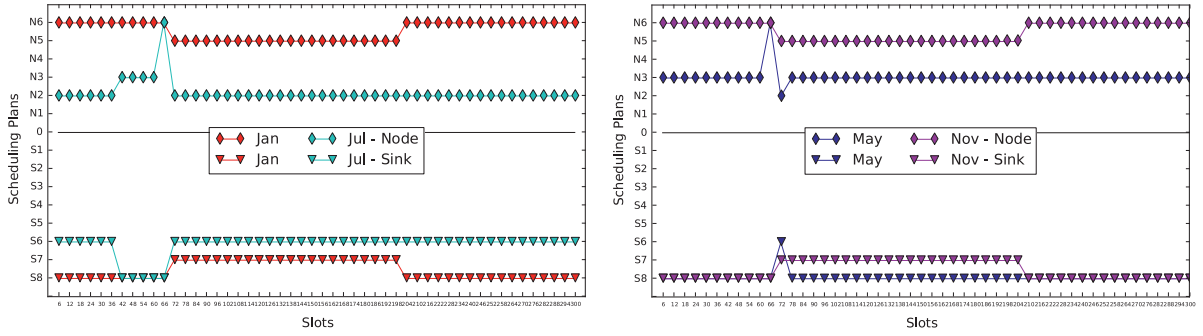


Fig. 5. Initial assignment of plans to slots for the months of January and July (on the left) and May and November (on the right) for a compatibility of at most 80%.

comparing the battery level achieved in the sink with 40% and 80% of compatibility: while the battery accumulated in the first case achieves 2030 mA h (see Fig. 4), in the second case is approximately 2022 mA h. Note that we are representing the execution during a day; the excess of production is accumulated along the time while the battery level is lower or equal than B_{max} ; otherwise, the battery is dissipated.

Fig. 5 shows the assignment of plans for the months of January and July (on the left) and for May and November (on the right) for the sensor and for the sink assuming at most 80% of compatibility. The axis y represents the plans in the sensor and in the sink and the axis x represents the slot where the plan is assigned. Remember that the order of the plans by efficiency is $\{N6, N5, N4, N3, N2, N1\}$ in the sensor and $\{S8, S7, S6, S5, S4, S3, S2, S1\}$ in the sink. Upon a change in the scheduling plan of the sensor, a change in the scheduling plan of the sink occurs unless the plan in the sink be still compatible with the new plan selected for the sensor. Consider, for example, the month of January where, as observed, the algorithm starts assigning the most efficient plan in the sensor (N6) and in the sink (S8). When the first change occurs in the sensor at slot 72 and the next plan N5 is selected, the sink looks for the first plan compatible with the new one. Since S8 is not compatible with N5, the sink selects the next compatible plan, which is S7, and starts to execute it at the same slot. When the sensor selects again N6 at slot 204, the sink comes back to S8. The same observation applies for the different months considered. For instance, in May, since N3 and N6 are compatible with S8, S8 is always selected to be executed. When the sensor changes to N2, since it is not compatible with S8 neither S7, the sink selects the next one compatible plan, which is S6.

We repeat the same experiments with a percentage of compatibility of at most 40%. Now, since we have more incompatible plans, we are increasing the probability of change in the sink, which looks for a plan less efficient but at the same time with a higher quality. Fig. 6 shows the results for the same months of the year. As a consequence of a lower compatibility, the sink assigns plans that are farther from S8 than in Fig. 5. In January for example, when N5 is selected, since N5 is not compatible with S7 and S8, the plan S6 must be selected. In July, the first compatible plan with N2 and N3 is S5. Similarly, in May, since N3 is executing in the sensor, S5 is selected. When the sensor downgrades and selects N6, the sink also downgrades and selects N8. As a consequence, this behavior trends to improve the overall quality of service of the assignment with regard to use a higher percentage of compatibility.

Fig. 7 shows the overall QoS achieved in the sensor and in the sink for compatibilities of 40% and 80%. QoS is computed as the average quality level of the plans assigned and therefore is higher in summer and lower in winter. We observe that the QoS in the sensor is larger than in the sink, due to the fact that our approach trends to select the highest quality plan as possible in the sensor and maintain the most efficient compatible plan in the sink. We also observe that the overall QoS with at

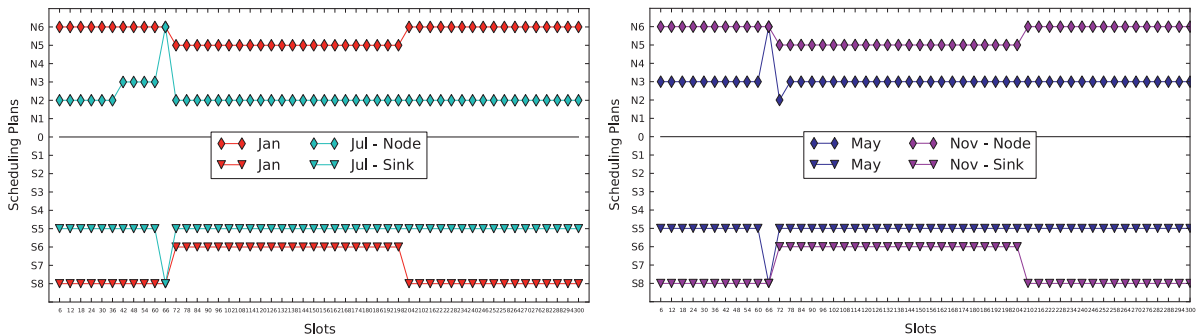


Fig. 6. Initial assignment of plans to slots for the months of January and July (on the left) and May and November (on the right) for a compatibility of at most 40%.

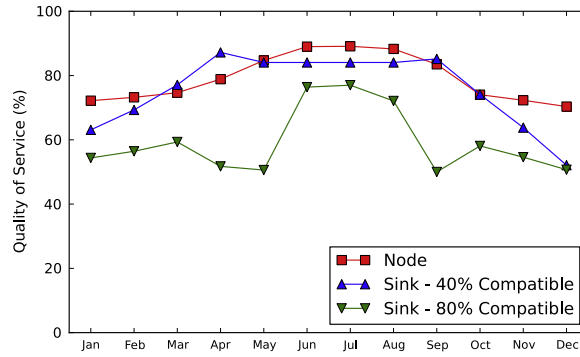


Fig. 7. Comparison of the overall quality of service for the sensor and for the sink.

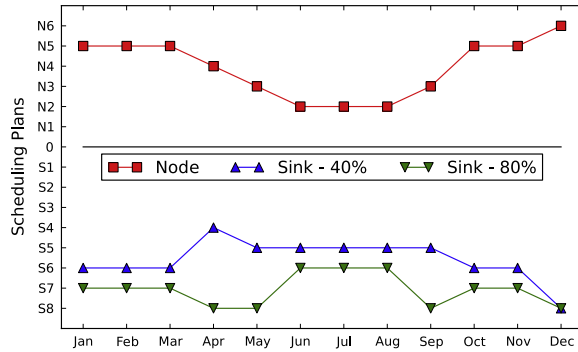
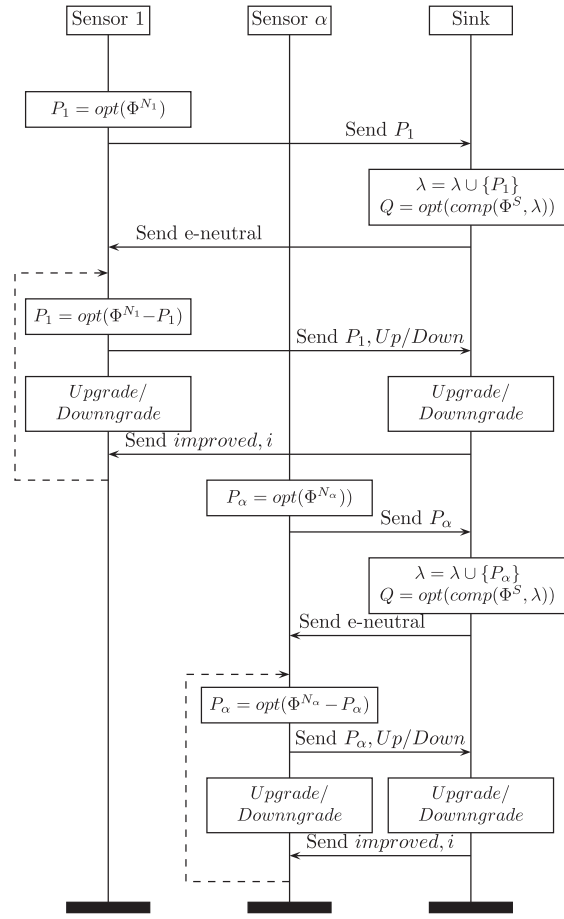


Fig. 8. Assignment in the sensor and in the sink at the slot of maximum production (12:00 am).

most 40% of compatible plans is higher than with a 80% of compatibility. For a compatibility of 40%, in the months of bigger production (from April to September) the QoS in the sink is similar to the achieved in the sensor. This happens because in these months the sensor upgrades several times to achieve the plan with the highest quality that is able to execute and, at the same time, with a low compatibility we reduce the possibility of that such a plan be compatible with the plan of the sink, forcing it to look for a plan with a higher quality, which, in turn, improves the overall QoS in the sink. However, with a higher compatibility, the QoS achieved in months of bigger production could be lower than in months of lower production. In particular, in the months of April, May, and September the overall QoS in the sink is close to 50%. This indicates that the most efficient plan, S8, is compatible with most of the plans selected in the sensor in these months, and therefore, S8 keeps running. To illustrate this, Fig. 8 shows the plans assigned in the sensor and in the sink at slot 150, which corresponds to the hour of maximum production (12:00 am). As described, the algorithm selects the most consuming plans in the sensor during the months of bigger production; however, the sink tends to maintain the most efficient plans while compatible with the selected in the sensor. Thus, for example, for a 80% of compatibility in April, May, and September, the sink conserves S8, while for a 40% the sink upgrades and selects S4 and S5 (see Table 3).

6. Generalization to an arbitrary number of sensors

The approach described so far considers two networked devices, 1 sensor and 1 sink. Let us consider now a more general scenario with α sensors ($\alpha > 1$) and one sink. Each sensor k ($1 \leq k \leq \alpha$) employs the energy harvesting system explained in Section 3.1 and runs the optimization algorithms described in Section 4. Our model extends naturally to α sensors by implementing independently the algorithms for each pair of devices $\langle k, \text{sink} \rangle$. This means that, any sensor k could select (independently from the others) the corresponding plan $P_i^{N_k}$ (i.e., the one that improves the solution or reduces the cost) among the set of their available scheduling plans Φ^{N_k} and, lately, communicate its choice to the sink. After receiving this message the sink proceeds as explained previously, computing the most efficient plan that is compatible with the plan received, i.e., $\text{opt}(\text{comp}(\Phi^S, P_i^{N_k}))$ and replying to the sensor the result of the evaluation of the energy neutrality condition (true or false). Both (the sensor and the sink) could enter in a loop (to improve the solution or to reduce the cost) that is executed a maximum number of n_k times, being n_k the number of scheduling plans of the sensor k . In this situation, a sensor l sends the plan selected $P_j^{N_l}$ to the sink. The sink then should compute the most efficient plan that is compatible with the two plans received from the sensors l and k , which would lead to two potentially different assignments of scheduling plans for the sink.

msc Communication Protocol α sensors-1 SinkFig. 9. Communication protocol between α sensors and 1 sink.

The plan to be selected for execution in the sink, let say Q , would be the most efficient plan that is compatible simultaneously with the plans $P_i^{N_k}$ and P_j . This process repeats each time that a sensor starts running the optimization algorithm by sending its plan to the sink. Thus, the maximum number of messages interchanged among α sensors and 1 sink considering one single execution of the algorithm is $\sum_{k=1}^{\alpha} 2(n_k + 1)$. It should be noted that, even considering different scheduling plans for different sensors, the transmission times should exhibit some relationship to ensure that they are compatible with the reception times in the sink. Note also that while this approach works for α sensors it could not deliver the optimal quality of service for the sink; we leave the optimization problem for α sensors as future work. Fig. 9 represents the communication protocol between the α sensors and the sink.

7. Conclusions

In this paper we address the problem of energy management affecting to two cooperating, networked devices, a sensor and a sink, both equipped with solar cells with the purpose of recharging their batteries to achieve potentially infinite life-times. We consider different scheduling plans for the sensor and for the sink, with different costs and quality levels, and a reference period of 24 h that is divided in time slots of the same duration. Based on this, we have proposed an algorithm to find a (sub)optimal assignment of scheduling plans to slots (for the node and for the sink) that aims at optimizing the overall QoS of the sensor and of the sink while keeping the system energy neutral, which means that the energy consumption in a reference period of time is never greater than the energy produced in the same period. We have also introduced the concept of compatibility between a couple of plans, one from the sensor and one from the sink, to express when two plans can run together, and which further restricts the set of plans to be selected.

We have evaluated the algorithm proposed by simulation. We take a representative configuration for the plans in the sensor and in the sink and we assume a low and a high compatibility between plans. Given such restrictions, the results reflect that our algorithm finds always an admissible solution (i.e. keep energy neutral the system) for every month of the year. The

algorithm also delivers a high QoS for the sensor close 90% in the months of summer, and close 70% in the months of December and January. We observe that the QoS achieved in the sink is dependent on the compatibility percentage: with a low compatibility the QoS is larger than using a high compatibility, due to the fact that the algorithm selects always the most efficient plan that is compatible with the plan in the sensor and, given a low compatibility, we are reducing the probability of that the plan in the sink be compatible with the selected plan in the sensor. Thus, while the QoS tends to be maximized in the sensor, we use a more conservative approach in the sink that tends to preserve a low cost. This may lead it to accumulate the energy produced but not consumed into the battery, which, in turn, could be further used to execute another tasks or attending a larger number of sensors. To this regard, although in this paper we have already presented the discussion of the generalization of the model to manage α sensors and 1 sink, we plan its optimization and its evaluation as future work.

Acknowledgments

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